

Edge AI SOP

Open the terminal/command terminal in Ubuntu and copy and paste a line of code into the terminal and press enter to execute the code - E.g 1. \$ cd Desktop

I. Install conversion environment

1. \$ cd Desktop
2. \$ sudo apt install python3-pip python3-venv build-essential libgraphviz-dev
3. \$ python3 -m venv hailo_env
4. \$ source hailo_env/bin/activate
5. \$ pip install --upgrade pip ; pip -V
6. Download: [hailo_dataflow_compiler-3.25.0-py3-none-linux_x86_64.whl](#)
7. Download: [hailo_model_zoo-2.9.0-py3-none-any.whl](#)
8. \$ cd ..
9. \$ mv Downloads/hailo_model_zoo-2.9.0-py3-none-any.whl Desktop/
10. \$ mv Downloads/hailo_dataflow_compiler-3.25.0-py3-none-linux_x86_64.whl Desktop/
11. \$ cd Desktop
12. \$ unzip hailo_dataflow_compiler-3.25.0-py3-none-linux_x86_64.whl
13. \$ pip install ./hailo_dataflow_compiler-3.25.0-py3-none-linux_x86_64.whl
14. \$ pip install ./hailo_model_zoo-2.9.0-py3-none-any.whl

II. Convert ONNX for hailo

1. \$ pip install notebook
2. Download: [All-processing-sample.ipynb](#) ; [Yolov5s_demo.onnx](#) ; [dataset.zip](#)
3. \$ mkdir notebooks
4. \$ cd ..
5. \$ mv Downloads/All-processing-sample.ipynb Desktop/notebooks
6. \$ mv Downloads/yolov5s_demo.onnx Desktop/notebooks
7. \$ mv Downloads/dataset.zip Desktop/notebooks
8. \$ cd Desktop/notebooks
9. \$ unzip dataset.zip
10. \$ jupyter notebook
11. Open: [All-processing-sample.ipynb](#) in Jupyter Notebook
12. Import Library: Click on the first block of code and Shift + Enter ([*] means waiting, [1] means done)
13. Set: [model_name](#) to the ONNX file name to yolov5s_demo
14. Use [Newtron](#) to check the start node and end node (Open the Yolov5 File)
15. Execute the code in each block once parameters are set(execute: Shift + Enter)
16. Copy the HEF file to the desktop

III. Inference in Hailo tappas

1. Download: [tonsofstreams.zip](#)
2. Extract the tonsofstreams.zip to the generalfolder, in the path
[Computer/Advantech/EdgeAISuite/Accelerator/Hailo_8/tool/tappas_v3.26.0/apps/h8/gstreamer/general](#)
3. Open the tonsofstreams folder in the terminal
4. \$ sudo chmod +x ./tonsofstreams.sh

5. \$./tonsofstreams.sh --num-of-sources 1 --num-of-devices 1
6. \$ sudo cp ~/Desktop/notebooks/yolov5s_demo/yolov5s_demo.hef ./
7. \$ sudo cp ../multistream_detection/resources/configs/yolov5.json ./
8. \$ sudo gedit ./yolov5.json
9. Copy the video you want the AI to detect into resources/byte_streams in the tonsofstreams folder
10. \$ sudo chmod +x *.sh
11. \$./convert_to_bytestream.sh <name of video to convert> bytestream0.mp4
12. \$ sudo gedit tonsofstreams.sh
13. Change line 14-16 to:

```
readonly POSTPROCESS_SO="$POSTPROCESS_DIR/libyolo_post.so"
readonly HEF_PATH="./yolov5s_demo.hef"
readonly DEFAULT_JSON_CONFIG_PATH="./yolov5s_demo.json"
```

14. \$./tonsofstreams.sh --num-of-sources 1 --num-of-devices 1
15. To change which video to use the AI model on, rename the file in byte_streams to bytestream0.mp4